

What is Structural Health Monitoring?

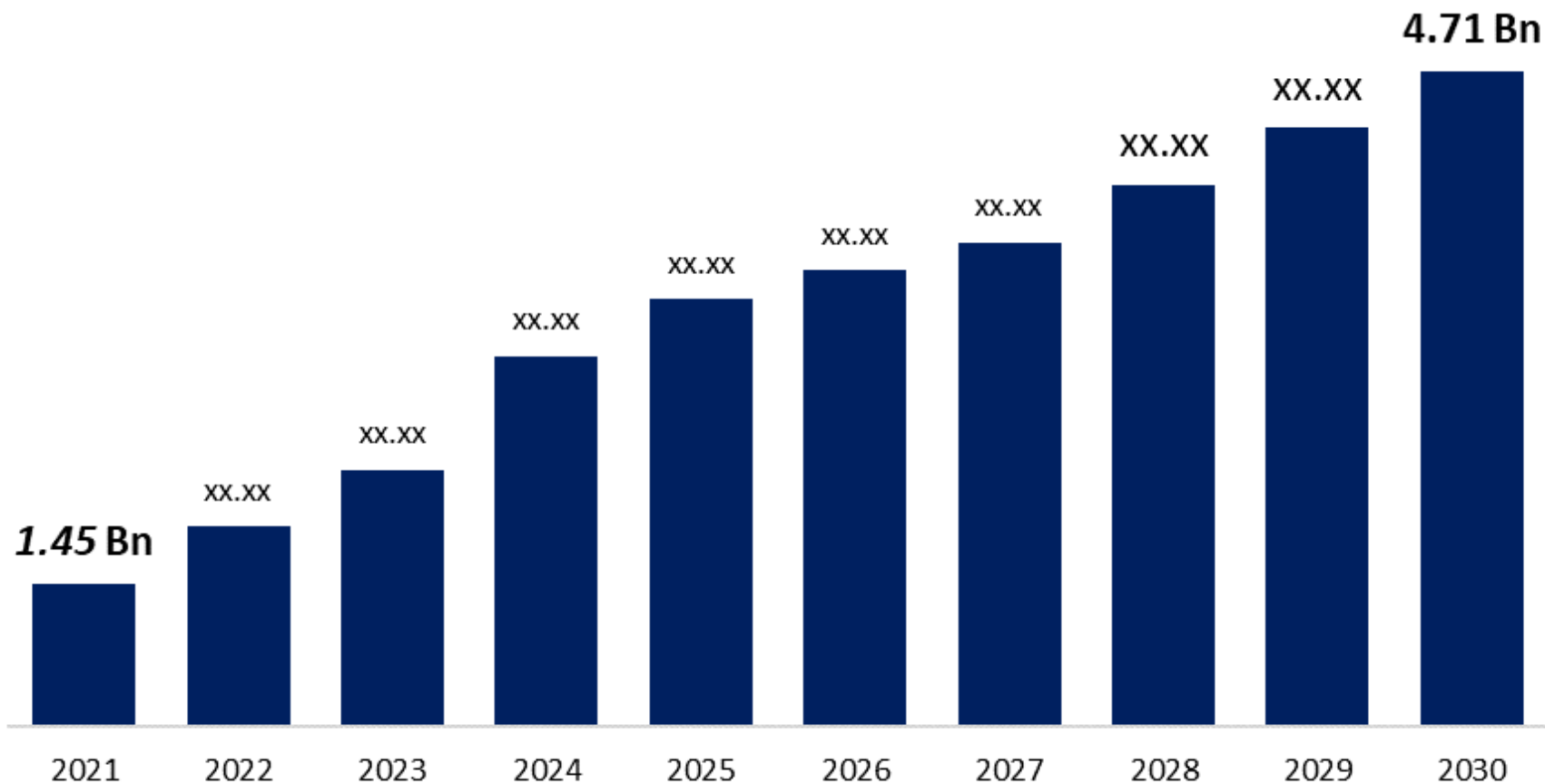
Why is it important?

Structural Health Monitoring (SHM) is a process of continuously or periodically monitoring the condition of a structure to assess its health and detect any potential damage, defects, or changes in its structural integrity. It involves the use of various sensors, data acquisition systems, and analysis techniques to monitor and evaluate the performance and safety of structures over time.

The main objectives of Structural Health Monitoring are:

- Early Detection of Damage
- Enhanced Safety and Reliability
- Cost-Effectiveness
- Real-Time Monitoring
- Continuous Assessment

Global Structural Health Monitoring Market





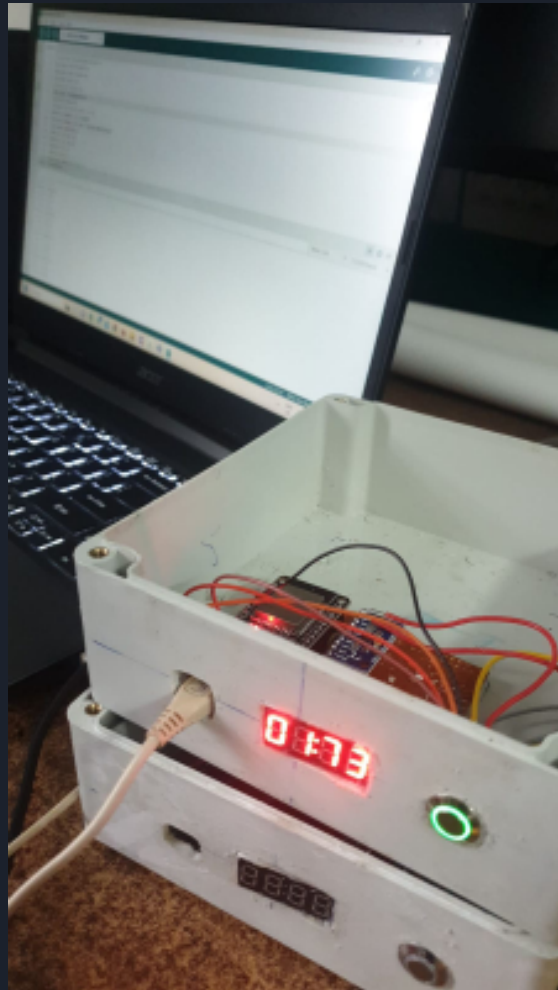
IoT-Based Tilt Meter: Revolutionizing Structural Insight

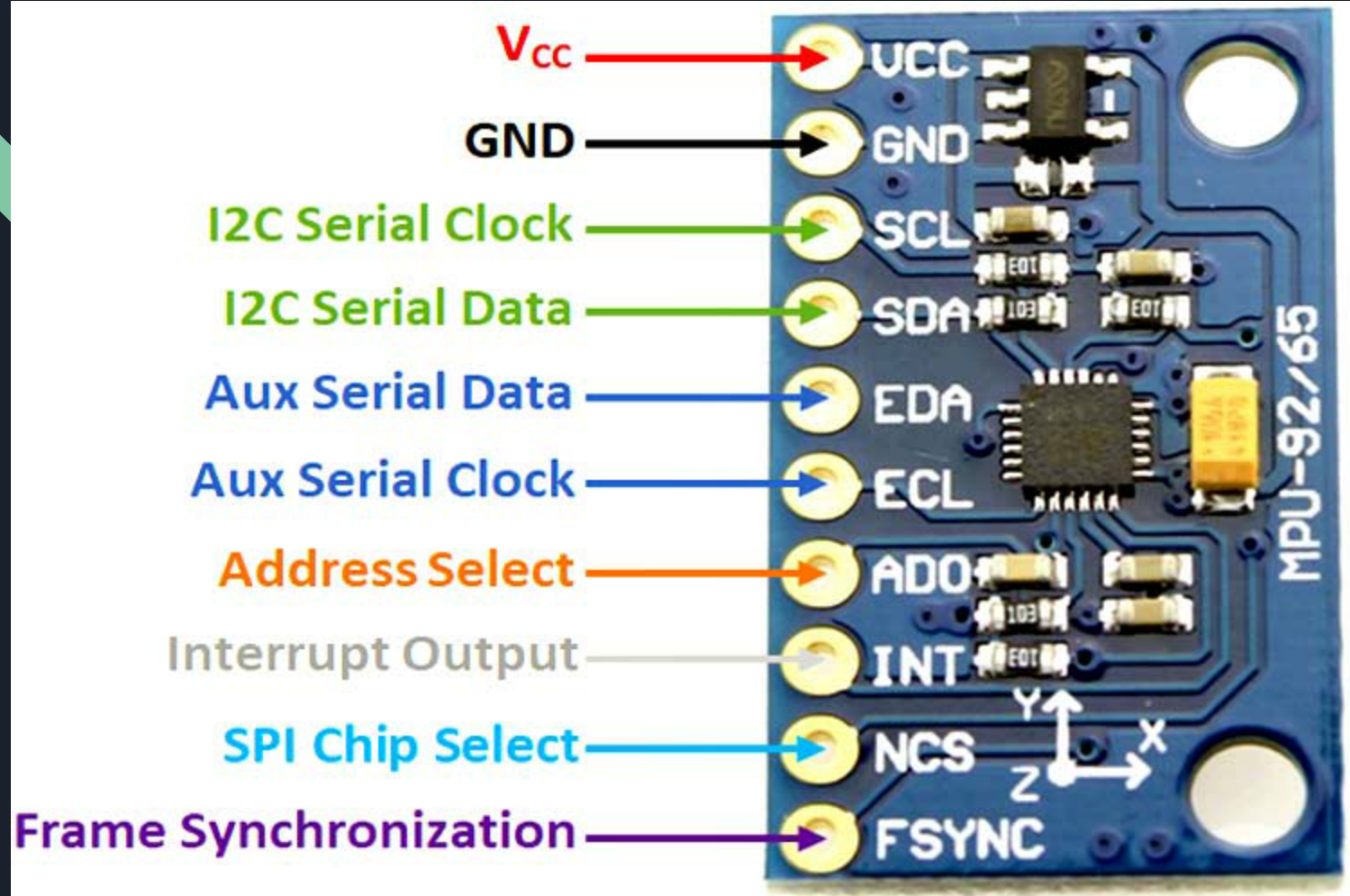
- Introducing our crown jewel: The IoT-Based Tilt Meter.
- Precision personified: Accurate monitoring of structural tilting angles.
- Enhancing data acquisition for comprehensive structural insights.
- A sentinel of stability: Safeguarding architectural treasures.
- Seamlessly integrating tradition with modern technology.
- Unveiling the mechanics behind our innovative tilt meter.
- Transforming tilting data into valuable structural knowledge.
- From craftsmanship to data-driven preservation.



Components and Technologies Used

1. MPU9250 Sensor
2. ESP32 WiFi Module
3. Arduino IDE
4. Reset Button
5. TM1637 Display
6. PCB
7. Thingspeak server
8. Blynk IOT
9. Wi-Fi Module: Enables data communication with the cloud



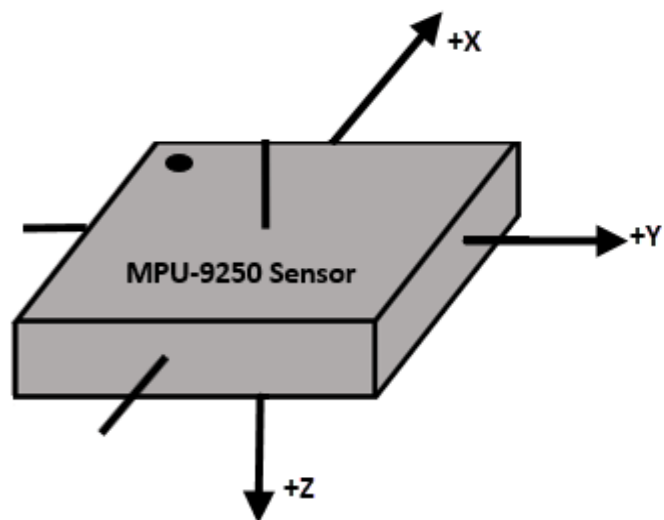




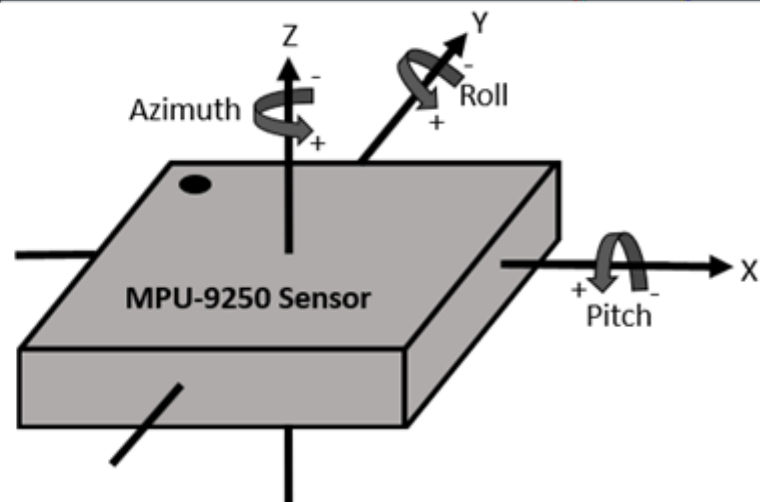
The MPU9250 integrates an **accelerometer**, a **gyroscope**, and a **magnetometer** into a single small package. The reason it is so small is that it uses Micro Electro Mechanical Systems (MEMS) – small mechanical components are made directly on the silicon die and are integrated with the electronic components to make a functional sensor.

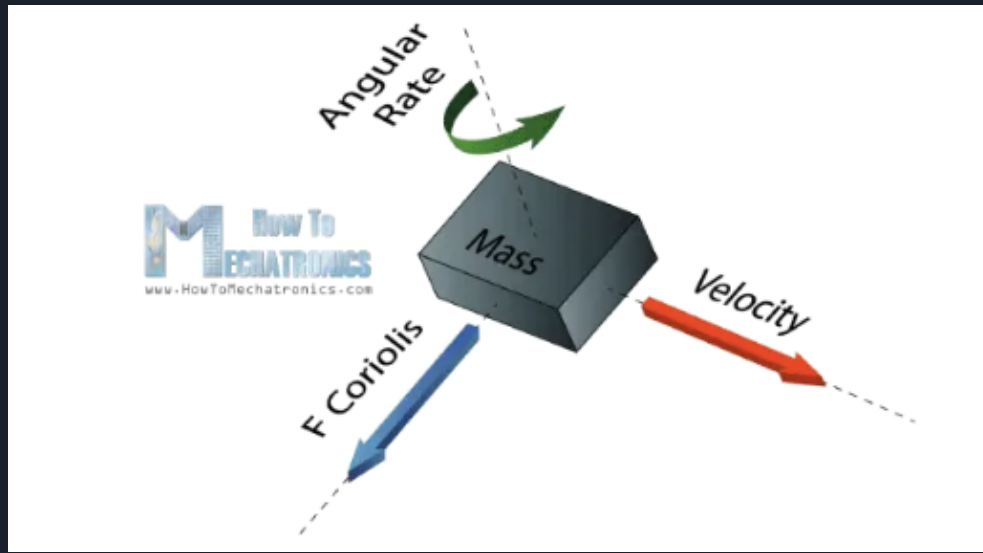
The module has a built-in LDO regulator, which supplied the MPU9250 with the 3.3V it needs. The LDO cannot handle very high voltages, and it is best powered with 5V. The module also has integrated pullup resistors for the I2C lines.

Orientation Axis for Magnetometer



Orientation Axis for Accelerometer and Gyroscope





The Gyroscope which measures rotational velocity/rate of change of angular position over time works through the Coriolis effect and MEMS technology.

When a mass is moving in a particular direction with a particular velocity and when an external angular rate will be applied as shown with the green arrow a force will occur which the sensor will detect.

As shown above, labelled by the blue and red arrow, this will cause perpendicular displacement of the mass. The displacement will cause a change in capacitance which will be measured, processed and it will correspond to a particular angular rate.

The outputs of the gyroscope are in degrees per second, so in order to get the angular position, we just need to integrate the angular velocity.



Project Designing

Define the requirements:

The tilt meter system will be used to monitor bridges, buildings, and other structures that are susceptible to damage from earthquakes, wind, or other environmental factors or any manmade causes.

Data will be collected every 10 seconds.

A level of accuracy of +/- 0.01 degrees is required.

Select the hardware and softwares:

The tilt meters is MPU9250 Sensor accelerometers.

The IoT gateway will be the ESP32.

The cloud storage platform we use is thingspeak..



Project Designing

Develop the software:

The firmware for the tilt meters will be developed using the Arduino IDE.

Test the system:

The hardware, software, and data collection and storage functions will be tested using a variety of tests, including unit tests, integration tests, and system tests.

Deploy the system:

The tilt meters, the IoT gateway, and the mobile app will be deployed to the target structures.



Working of the IoT-Based Tilt Meter

1. Tilt Sensor:

Measures precise tilting angles of the structure.

Converts physical tilt into electrical signals.

2. Microcontroller:

Processes incoming sensor data.

Performs necessary calculations and analysis.

3. Wi-Fi Module:

Establishes a seamless connection to the cloud.

Facilitates real-time data transmission.



Working of the IoT-Based Tilt Meter

4. ThingSpeak Cloud Platform:

- Securely stores and manages tilt data.

- Enables remote access and retrieval.

5. User-Friendly Mobile App:

- Provides intuitive interface for users.

- Allows real-time monitoring and analysis.

6. Data Visualization :

- Generates dynamic graphs of tilt angles over time.

- Offers insights into structural behavior trends.



X TILTsensor

ooo



Angle is deflected more than enough Inbox



Blynk 9:51 pm
to me ▾



Take care of ur sensor



Blynk 9:56 pm
to me ▾



Take care of ur sensor



← Reply

↩ Reply all

→ Forward

Request timeout..

x_axis = 6.04

y_axis = 47.93

z_axis = 7.36

Request timeout..

x_axis = 6.15

y_axis = 47.69

z_axis = 7.55

Request timeout..

x_axis = 5.89

y_axis = 47.46

z_axis = 7.26

Request timeout..

x_axis = 0.00

y_axis = 0.00

z_axis = 0.00

Request timeout..

x_axis = 0.37

y_axis = -0.23

z_axis = -0.05

Request timeout..

x_axis = -0.45

y_axis = -0.85

z_axis = -0.19

Request timeout..

x_axis = -0.10

y_axis = -0.35

z_axis = -0.10

Request timeout..

x_axis = 0.00

y_axis = -0.31

z_axis = -0.19



Future Scope and Beyond

1. Advanced Analytics:

Implement machine learning algorithms for predictive analysis.

Detect early signs of structural deterioration based on historical data.

2. Multi-Sensor Integration:

Integrate additional sensors for comprehensive structural insights.

Combine tilt data with temperature, humidity, and stress sensors.

3. Remote Diagnostics:

Develop remote diagnostic capabilities for real-time troubleshooting.

Enable remote adjustments and calibrations for optimal performance.

4. Smart Alert System:

Create an intelligent alert system for critical tilt deviations.

Notify stakeholders instantly for immediate action.



Future Scope and Beyond

5. Integration with Building Management Systems:

Seamlessly integrate tilt data into existing building management systems.

Enable holistic management and monitoring of structures.

6. Research Collaborations:

Partner with academic institutions for further research and development.

Explore new applications and innovations to improve efficiency and inculcate new ideas for even better structural health monitoring

7. Global Implementation:

Expand deployment of our IoT-Based Tilt Meter worldwide.

Contribute to global structural health monitoring and preservation.



Conclusion

1. The IoT-based tilt meter system was successfully designed, developed, and implemented.
2. The system is able to monitor the tilting angles of structures in real time with a high level of accuracy.
3. The system has the potential to improve the safety of structures by providing early warning of potential damage.
4. The system is cost-effective and easy to deploy.
5. The system is scalable and can be used to monitor a wide variety of structures.



Reference

- [IoT-Based Tilt Meter Systems for Structural Health Monitoring by Mohamed Abdelaziz](#)
- [Structural Health Monitoring with Wireless Sensor Networks by A.M. Sayed, A.A. Awad, and M.G. Abdel-Rahman.](#)
- [A Review of IoT-Based Tilt Meter Systems for Structural Health Monitoring](#)
- [Design and Implementation of an IoT-Based Tilt Meter System for Structural Health Monitoring](#)

Thank you!

